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Problem Definition

Low Earth Orbit (LEO) satellites need to be able to serve traffic demand that is both constantly changing, and spread unevenly over a huge area, with the limited number of resources available to them. Resource allocation is especially difficult to do efficiently as delivering traffic to a geographic cell is heavily constrained due to network capacity. Satellites need to have enough radio frequency (RF) capacity to support the cell, ground stations must have necessary capacity to transport the traffic between the network of satellites and the infrastructure on the ground, and finally links between satellites need to have the necessary bandwidth available. What makes efficient allocation even harder is the fact that all these resources are interconnected: for example allocating more RF capacity to an area with high demand may not actually increase throughput if the links between satellites are already congested.

Thus, our project aims to address the problem of dynamically allocating these resources while accounting for the constraint of how these resources are dependent on each other. Specifically, we plan to focus on three resources: RF capacity for every satellite, capacity for links between satellites, and capacity of the ground stations. Our aim is to develop a mechanism that can distribute RF capacity among geographic cells based on demand, congestion, and capacity available downstream while also accounting for how to be fair to regions that have received insufficient service in the past. Rather than optimizing RF allocation independently, this mechanism will take into account whether traffic assigned to a cell can be carried by the satellite network to an available ground station. We will also be using COSMOSim to investigate whether coordinating these 3 resources actually improves fairness, demand satisfaction, and end-to-end throughput compared to other less comprehensive policies.

Motivation

LEO satellite networks are becoming more and more important for ensuring high bandwidth, low latency connectivity is available in remote regions where ground infrastructure is often unavailable or infeasible to build. In such regions, demand is usually sporadic with some pockets experiencing high congestion while others have little to no demand at all. Static resource allocation policies waste the limited satellite capacity available by trying to allocate resources to low-demand pockets while the highly congested pockets are left underserved. Demand-based allocation is better at addressing this imbalance but allocating resources based only on demand has not guaranteed better end-to-end performance (based on past studies). For example, if we assign a greater number of RF channels to a highly congested pocket, the benefit we get may be much less than expected as this traffic may still have to go through an already overloaded

inter-satellite link. Thus, when making such resource allocation decisions it is important to consider the capacity limitations present network-wide rather than just trying to optimize a resource independently.

Additionally, resource allocation needs to be done in a way that balances the overall performance of the network while also ensuring all parts of the region are served fairly. Policies that only focus on maximizing throughput might give higher priority to cells that have better network conditions and less congested paths while leaving more remote or more congested regions with less service. On the other hand, policies that only focus on fairness might reduce the overall network throughput by allocating too many resources to a region that has very limited downstream capacity. Balancing these tradeoffs only becomes more difficult when accounting for satellite movement, changes in network connectivity, and changes in demand over time. Thus, by considering the allocation of RF capacity, inter-satellite link congestion, and bandwidth of ground stations, we aim for our proposed mechanism to be able to use the available resources more effectively to ensure that service is improved in underserved areas. By evaluating this, we will be able to further understand when such multi-resource allocation performs better than simpler policies such as equal or demand-proportional allocation.

Challenges

1. Tight resource coupling

The primary challenge when handling dynamic resource allocation for LEO satellites is the tight coupling between available resources. Adding additional RF channels to a cell doesn't necessarily increase its useful throughput, since the serving satellite must also have uncongested ISL paths to a ground station gateway with enough feeder link capacity to serve the traffic. Optimizing only one part of the path might just move the bottleneck to a different place. For example, a demand-aware RF policy might allocate more channels to an already congested cell, leaving that newly assigned RF capacity underutilized.

2. Changing topology and demand

LEO mobility introduces another challenge. As the satellites move, their visibility to cells and gateways change, and the available paths through the constellation may change as well. An efficient allocation at one moment in time may quickly become insufficient or impossible shortly afterwards. The allocator must be able to respond to these changes in real time without excessively reassigning beams and routes, which could disrupt the network. Similarly, traffic demand is constantly changing and varies by region. Optimizing purely for total throughput prioritizes cells with strong access links and short, uncongested paths, leading to starvation for cells in remote or poorly connected areas. Conversely, overdoing fairness may allocate too many resources to cells with limited downstream capacity and heavily reduce network throughput. The

allocation policy must take into account the changing topology and dynamic, uneven demand to ensure efficiency and fairness.

3. Computational constraints

Finally, computing an exact global optimal decision at every interval would likely be too expensive under real-world time constraints for a large constellation. We would instead have to develop a scalable, “close to optimal” algorithm to determine how much added complexity returns meaningful improvements to network throughput and fairness.

Methodology

General Setup

We will use COSMOSim as the foundation for simulating a Low Earth Orbit (LEO) satellite network. We will build on its existing capabilities to generate time-indexed network graphs, represent satellite connectivity, distribute user terminals across geographic cells, generate traffic demands, and calculate routed capacities.

The simulated network will include satellites, geographic cells, user terminals, inter-satellite links (ISLs), and ground station gateways. Since satellites move over time, the available connections between satellites and cells may change between graph snapshots. Our algorithm will account for these changes when determining how to allocate resources.

We will focus on three main resource dimensions:

1. **Satellite RF Capacity:** The limited radio resources available for satellites to serve geographic cells. This will be the primary resource our algorithm actively allocates.
2. **ISL Capacity:** The capacity of links connecting satellites, which limits how much traffic can be forwarded through the constellation.
3. **Gateway Capacity:** The capacity of ground stations connecting the satellite network to terrestrial networks, which limits how much traffic can leave the satellite network.

We will also account for terminal distribution and per-cell capacity constraints. Terminal distribution determines where traffic demand originates, while per-cell capacity limits how much traffic a cell can receive or use. These factors will help determine where additional RF resources are needed and whether allocating more resources would allow additional traffic to be delivered.

We will first examine COSMOSim's existing allocation, routing, and capacity-calculation components to determine which can be reused and which require modification. This will allow

us to build on the existing simulator while developing the additional logic needed to implement our algorithm.

Development of the Resource Allocation Algorithm

We will modify COSMOSim's existing beam and channel allocation policies to implement our dynamic resource allocation algorithm. At each allocation period, the algorithm will evaluate traffic demand across geographic cells and determine how to distribute available RF resources. Unlike an approach based on demand alone, our algorithm will also estimate the capacity available along routes to ground station gateways and account for congestion that could prevent allocated RF resources from translating into delivered traffic.

For example, a geographic cell may have high traffic demand and need additional RF capacity, but the ISL along its route to a gateway may have little remaining capacity. Allocating more RF resources to that cell would not necessarily increase the amount of traffic delivered because the ISL would remain a bottleneck. Our algorithm will account for this limitation when assigning RF resources and adjust allocation priorities to consider both unmet demand and downstream network capacity. This approach is intended to prevent resources from being allocated based on demand alone when the network cannot carry the resulting traffic.

At each allocation period or graph snapshot, our algorithm will follow these steps:

1. **Measure cell demand and fairness.** For each geographic cell, determine its traffic demand and the amount of traffic it currently receives. Compare demand with delivered traffic to identify underserved cells. This information will contribute to allocation priorities so that cells receiving less service than they need are considered alongside cells with high demand.
2. **Estimate available network capacity.** For each eligible cell-satellite pair, identify a potential route to a ground station gateway and estimate how much additional traffic the route can carry. This estimate will consider the available RF capacity, the remaining capacity of the ISLs along the route, and the remaining capacity of the gateway and ground link. For a single route, the most restrictive component will limit the additional traffic that can be delivered. We will build on COSMOSim's existing routing and capacity-calculation functionality, adding logic to estimate remaining capacity where necessary.
3. **Allocate RF channels and beams.** Determine which geographic cells should receive RF resources based on their unmet demand, fairness priority, and estimated downstream capacity. Assign available beams and channels while respecting satellite RF capacity limits, satellite coverage, and COSMOSim's interference restrictions. Where per-cell capacity limits apply, allocations will also respect those limits. Cells whose routes have

insufficient remaining ISL or gateway capacity will receive lower priority for additional RF resources because those resources may not increase the amount of traffic delivered.

4. **Calculate deliverable capacity.** After allocating RF resources, use COSMOSim's routing and capacity-calculation workflow to estimate how much traffic can be carried through the network to ground station gateways. Compare this estimate with cell demand to identify remaining unmet demand and determine whether ISL or gateway capacity is limiting traffic delivery. We will connect this calculation to the allocation process as needed so that changes to RF assignments are reflected in subsequent capacity estimates.
5. **Update congestion weights and allocation priorities.** Calculate ISL and gateway utilization relative to their respective capacities. Increase the congestion penalty for links and gateways with less remaining capacity, then use these penalties when ranking subsequent allocation decisions. This will favor alternative routes or allocations that place less pressure on congested network components when alternatives are available. We will also incorporate fairness priorities to account for cells that continue to receive less service than their demand requires.
6. **Repeat the process as the network changes.** Apply the allocation process across successive network snapshots as satellite connectivity and traffic demand change. At each new allocation period, update demand, available capacity, and congestion estimates before determining the next allocation. This will allow the algorithm to respond to changing network conditions rather than relying on a fixed allocation.

We will begin by running COSMOSim's existing allocation and capacity-calculation workflows to understand how RF allocation, routing, and network capacity are represented. We will then modify the RF allocation policies to incorporate downstream ISL and gateway capacity estimates, adding logic to adjust allocation priorities based on congestion and fairness. We will also connect the allocation and capacity-calculation stages as needed so that the effects of allocation changes can inform subsequent decisions. The specific scenarios, baselines, and performance measures used to evaluate our approach are described in the following Evaluation Plan section.

Related Works

Previous work on LEO resource allocation has really mainly focused on the radio link between a satellite and the cells that it serves. The following three studies are closest to what our project aims to accomplish.

Guo et al.

This paper treats beam hopping as a way to improve the resource utilization by dynamically allocating time, power, and frequency according to the traffic requests. Their mechanism also satisfies GEO interference avoidance. They model the joint allocation of time slot, frequency,

and power in a multi beam LEO system as a multi-objective optimization problem, and report that the gains are largest when traffic demand is unevenly distributed. This supports our idea that demand aware allocation beats static allocation under skewed demand. However, the optimization stops at the satellite to cell RFC link. It doesn't ask whether the traffic admitted to a cell can actually be carried to a ground station. Our project uses a similar demand aware RF allocation step, but each beam or channel decision also depends on the ISL and gateway capacity reachable from that satellite.

Zhao et al.

This paper extends demand aware beam hopping to multiple satellites. They study a digital twin based collaborative allocation network for multiple LEO satellites with overlapping coverage areas. The digital twin later designs each satellite's beam hopping pattern to allocate overlapping cells and the LEO layer optimizes power for each selected cell. Beam patterns are learned with Actor-Critic networks and power allocation uses multi-agent reinforcement learning with each beam as an agent. What's interesting is that the authors report about a 72.5% reduction in satellite load disparity and an average delay of 12ms. This is the closest prior work to ours by far because it balances load across satellites rather than within one. Its notion of load is still limited to satellite access side queues. It doesn't model inter-satellite routing or feeder-link limits and its objective is load balancing across satellites rather than fairness across cells over time. The utility of our work is that it adds both in this regard. It also adds a fairness term that raises the priority of cells that have been recently underserved.

Lu et al.

This paper proposes a downlink optimization algorithm that adapts resource allocation to changing link conditions in LEO networks. Their work shows the value of reacting to time-varying channel quality. However in this case it optimizes for a single factor which is the state of the access link. Our project keeps link-condition awareness as one input. We combine it with demand, fairness, and congestion on the ISL and gateway layers, because a strong access link has limited value if the backhaul path behind it is saturated.

All three works optimize RF resources on the satellite cell to cell link. Each evaluates against its own simplified simulator. Our project builds on their demand aware beam and channel prioritization but treats RF capacity, ISL capacity, and gateway feeder capacity as doubled resources. Allocating any one of them in isolation can simply move the bottleneck elsewhere. We also add an explicitly fairness mechanism based on recent service history. Finally, we evaluate COSMOSim's Starlink-scale constellation which has time varying visibility and topology, and use ablations that isolate the contribution of ISL awareness, gateway awareness, and fairness.

References:

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Evaluation Plan

We will evaluate whether our proposed allocation policy improves end-to-end service by considering satellite RF capacity, inter-satellite link (ISL) capacity, and gateway capacity together. Using COSMOSim, we will compare allocation methods under identical network conditions and examine their tradeoffs between throughput, demand satisfaction, fairness, and resource utilization.

We will test each allocation method under every test scenario and collect the same evaluation metrics for each run, where supported by COSMOSim or calculated using its outputs. This will allow us to compare the methods under identical conditions and determine how their performance changes across different network conditions.

1. Baseline Methods

We will compare our approach against two primary baselines:

- **Method 1 Equal RF allocation:** Available RF resources are divided equally among eligible cells, subject to satellite capacity and visibility constraints. This provides a simple reference for determining whether adapting allocation to demand and network congestion improves performance. We will verify whether COSMOSim already supports this allocation strategy or implement it if necessary.
- **Method 2 Demand-proportional RF allocation:** Available RF resources are distributed in proportion to each eligible cell's current traffic demand, without considering downstream ISL or gateway congestion. This comparison will show whether accounting for downstream bottlenecks provides benefits beyond demand awareness alone.

2. Algorithm-Specific Methods

We will also compare simplified versions of our proposed policy to isolate the contributions of downstream capacity awareness and fairness.

- **Method 3 RF allocation without ISL awareness:** The algorithm will consider demand, gateway capacity, and fairness but will not account for ISL congestion when determining allocation priorities. This comparison will help establish whether ISL awareness improves performance when inter-satellite links become bottlenecks.
- **Method 4 RF and ISL coordination without gateway awareness:** The algorithm will account for ISL capacity but not gateway capacity. This will help establish whether gateway awareness provides additional benefits beyond ISL coordination.
- **Method 5 Full coordination without fairness terms:** The algorithm will retain RF, ISL, and gateway awareness but will not prioritize underserved cells based on fairness. This comparison will help determine whether fairness priorities reduce starvation and improve the distribution of service, and whether these benefits involve tradeoffs in overall capacity.

2. Test Scenarios

We will test each method under four main scenarios:

- **Uniform demand:** Cells have similar traffic demand. This establishes performance under balanced conditions and checks whether the proposed policy introduces unnecessary overhead or uneven service.
- **Demand hotspots:** A subset of cells has substantially higher demand than surrounding cells. This tests whether the policy can direct resources toward congested areas while maintaining service for other cells. We will also shift hotspot locations over time to evaluate adaptation.
- **Limited ISL capacity:** We will reduce ISL capacity while holding RF and gateway capacities fixed. We will vary the degree of capacity reduction to examine different levels of congestion. This tests whether the policy avoids assigning RF resources to cells whose traffic cannot be carried through the satellite network.
- **Limited gateway capacity:** We will reduce gateway or feeder-link capacity while holding RF and ISL capacities fixed. We will vary gateway capacity to examine different levels of congestion. This tests whether gateway awareness improves service when access to ground infrastructure becomes the bottleneck.
- **Combined ISL and gateway constraints:** Where feasible, we will constrain both ISL and gateway capacity to examine how the algorithm performs when multiple network resources limit traffic delivery.

We will vary the offered traffic load and the constrained resource's capacity to examine both lightly loaded and congested conditions. If time and available input data permit, we may also test

selected COSMOSim-supported geographic regions with different geographic and connectivity characteristics to examine whether our results generalize beyond a single network configuration.

3. Evaluation Metrics

We will measure:

- **End-to-end throughput:** Traffic successfully delivered through the network, rather than traffic assigned RF capacity alone.
- **Demand satisfaction:** The fraction of offered demand successfully served, reported both overall and per cell.
- **Fairness and starvation:** Jain's fairness index calculated over per-cell demand satisfaction, together with the proportion of cells with positive demand that receive no service during each measurement interval.
- **Resource utilization:** Used capacity relative to available capacity for RF resources, ISLs, and gateways.
- **Latency and congestion:** End-to-end delay and queue buildup where supported by the simulation.
- **Runtime:** The time required to compute allocation decisions at each simulation step.

4. Experimental Procedure and Success Criteria

All methods will use the same constellation, visibility timeline, terminal distribution, traffic inputs, and capacity limits. Routing will remain consistent across the primary comparisons to isolate the effect of allocation decisions. For scenarios with randomly generated demand, we will repeat experiments with multiple seeds and report average results and variability.

We will collect the evaluation metrics for every method under every scenario and compare the results to identify which methods perform best under different network conditions.

We will consider the approach successful if resource awareness improves delivered throughput or demand satisfaction under downstream bottlenecks, while fairness terms reduce starvation and improve the distribution of service. We will explicitly report any capacity–fairness tradeoffs and computational overhead, including conditions where simpler baselines perform equally well.